to that API. The entire process of setting up infrastructure, installing software, and building clusters from the hardware layer to API availability is unique to edge deployments, and a fundamental challenge compared to cloud deployments where everything is already managed for you.

Q. How does SUSE navigate through these challenges?

A. We have software solutions to address these challenges, offering three provisioning models depending on the use case. For mass onboarding, we provide a tool called Edge Image Builder, which allows you to create a customised onboarding image. This image can be burned onto any hardware, shipped, or loaded onto a USB stick, significantly speeding up the process of taking a machine from an off state to a fully running cluster. It handles everything, including phoning home capabilities, security, and other essential configurations. Additionally, we offer adjacent tools that help manage the stack and solve both Day Zero and Day One challenges. These solutions are part of our comprehensive offerings.

Q. What kind of business model does SUSE follow?

A. We monetise the support for the free and open source software. While the technology stack of Linux is very mature, Kubernetes operates on a much faster release cycle, with new versions coming out every three months. We reconcile these differing life cycles between Linux and Kubernetes, with seamless upgrades for Kubernetes while keeping the entire stack stable. We have validated designs and a CI/CD pipeline that runs thorough validation checks to ensure that when Kubernetes is upgraded your applications and infrastructure remain intact and functional.

Our support services help customers maintain smooth operations while dealing with the frequent updates in Kubernetes and slower-moving Linux base layers.

Q. Do you also incorporate the help of the community in offering these support services?

A. Most of our support comes directly from SUSE employees, who contribute upstream and work closely with the community. It's very rare that we encounter an issue we can't resolve ourselves, requiring us to reach back to the wider community for help. In most cases, our engineers are the ones who wrote the code, so customers often get direct support from the people who developed the software.

Q. In India, which industry segments are you targeting with your product for the edge?

A. The three primary industries we've identified so far are telecom, manufacturing, and banking. While we're still evaluating, there may also be opportunities in healthcare.

Q. What products do you have for the India market?

A. SUSE offers two key products for the India market. The first is SUSE Edge, which includes all the necessary components for onboarding, setting up, and configuring Kubernetes, supporting multiple architectures.

The second product is the Adaptive Telco Infrastructure Platform (ATIP), which is optimised for the high-performance requirements of telco customers. ATIP is a commercial implementation of the Sylva specification, tailored for 5G deployment. This telco-focused product supports various hardware types and includes optimisations like real-time kernel implementation, Precision Time Protocol (PTP), Single Root I/O Virtualization (SR-IOV), and Data Plane Development Kit (DPDK) to efficiently move packets from network interface cards (NICs) to containers.

Q. What's your strategy for the Indian market?

A. We've discovered that in India, customers and CIOs are seeking a complete end-to-end solution—from the bottom of the stack to full application support. To effectively meet this demand, we are partnering with integrators, systems integrators, and value-added resellers who can help deliver these comprehensive solutions.

Q. How does the deployment of the Kubernetes platform differ across various applications?

A. We have three large deployment methodologies that include all the bare metal management capabilities and cover the edge as well. At SUSE, we manage hardware using BMC (baseboard management controller) interfaces, such as ILO, DRAC, or IPMI, which are the out-of-band management interfaces for data centre-grade hardware.

For instance, if you have five racks of gear, we can spin up two or three of those racks at any time to build clusters from their resources. When demand increases, additional racks can be powered on, providing extra capacity. Conversely, when demand decreases, machines can be powered down, spinning down the cluster. This is akin to the elasticity offered by cloud service providers, but instead, it's hardware elasticity that optimises energy usage and contributes to green computing.

The third and more simple deployment method involves shipping hardware or building an image such that when a machine is powered on, the OS boots up, connects to Rancher, and registers for control from there. The deployment infrastructure is different in that case.

Q. What extensions are required when we integrate different edge devices across the Kubernetes platform, and how do they work?